

All-Orders Asymptotic Transseries for the Swing Factorial and Its Two Inverse Branches

Bernoulli–Bell coefficient calculus, Lambert– W normal forms,
logarithmic reversion, and beyond-all-orders control

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Abstract

The swing factorial, OEIS A056040,

$$\text{sw}(n) = \frac{n!}{[n/2]!^2},$$

is not asymptotically represented by one smooth branch: its even and odd subsequences have different power prefactors and the full sequence repeatedly rises and falls. We therefore split it into the two gamma-ratio interpolants

$$S_\varepsilon(m) = \frac{\Gamma(2m + \varepsilon + 1)}{\Gamma(m + 1)^2}, \quad \varepsilon \in \{0, 1\},$$

so that $S_\varepsilon(m) = \text{sw}(2m + \varepsilon)$ for integral $m \geq 0$. For each branch we derive the complete Poincaré transseries

$$S_\varepsilon(m) \sim K_\varepsilon 4^m m^{\beta_\varepsilon} \exp\left(\sum_{r \geq 1} \lambda_r^{(\varepsilon)} m^{-r}\right) = K_\varepsilon 4^m m^{\beta_\varepsilon} \sum_{r \geq 0} c_r^{(\varepsilon)} m^{-r},$$

where $K_\varepsilon = 2^\varepsilon / \sqrt{\pi}$, $\beta_\varepsilon = \varepsilon - \frac{1}{2}$, the logarithmic coefficients are explicit Bernoulli-polynomial differences, and the multiplicative coefficients are complete Bell polynomials. We give partition sums, triangular recurrences, parity-transfer identities, exact coefficient tables, rigorous fixed-order remainders, and an exact Binet-integral completion.

Each S_ε is strictly increasing on the relevant half-line and hence has a real inverse M_ε . Its dominant power–exponential phase is inverted exactly by a Lambert function: W_{-1} for the even sheet and W_0 for the odd sheet. Writing that Lambert core as $w_\varepsilon(y)$, we prove the all-orders inverse expansion

$$M_\varepsilon(y) \sim w_\varepsilon(y) + \sum_{r \geq 1} d_r^{(\varepsilon)} w_\varepsilon(y)^{-r}.$$

The coefficients $d_r^{(\varepsilon)}$ are specified in three equivalent ways: a triangular coefficient-extraction recurrence, a closed Lagrange–Bürmann differential formula, and a finite coefficient-array algorithm. Finally, we unfold the Lambert core into the ordinary polynomial–logarithmic scale

$$M_\varepsilon(y) \sim \frac{1}{\log 4} \left(X_\varepsilon - \beta_\varepsilon \log X_\varepsilon + \sum_{r \geq 1} \frac{P_r^{(\varepsilon)}(\log X_\varepsilon)}{X_\varepsilon^r} \right),$$

with an all-orders recurrence for every coefficient of every polynomial $P_r^{(\varepsilon)}$. Numerical residual tests confirm the formulae to the displayed orders.

Interpretation of “full transseries”

The word *full* is used in three precise senses here: all algebraic orders are given by explicit coefficient formulae; both parity sheets are included; and the exact Binet integral is retained as the canonical beyond-all-orders completion. A further decomposition of the optimally truncated Binet remainder into terminant/Stokes sectors is representation-dependent. The transport rule for any such exponentially improved sector is nevertheless derived in [Section 8](#).

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1 Introduction and statement of the main results

The swing factorial appears deceptively close to the central binomial coefficient, yet its parity structure changes the inverse problem qualitatively. At even arguments it is the central binomial coefficient; at odd arguments it is that coefficient multiplied by a linear factor. Thus the direct asymptotics require a period-two transseries, while an inverse of the entire sequence does not exist in the usual single-valued sense.

The derivation has four layers.

1. The duplication formula reduces both parity branches to a common gamma ratio.
2. Shifted Stirling expansions express the logarithmic coefficients as Bernoulli-polynomial differences. Exponentiation is then handled exactly by complete Bell polynomials.
3. The leading inverse phase $\alpha m + \beta \log m$ is solved exactly with a branch of Lambert W .
4. Lagrange–Bürmann inversion around that exact carrier produces the complete inverse-power correction grid. A second reversion unfolds W into powers of $1/\log y$ with polynomial coefficients in $\log \log y$.

The two carrier constants used throughout are

$$\alpha = \log 4, \quad K_\varepsilon = \frac{2^\varepsilon}{\sqrt{\pi}}, \quad \beta_\varepsilon = \varepsilon - \frac{1}{2}, \quad \varepsilon \in \{0, 1\}. \quad (1)$$

Main direct and inverse formulae

For $m \rightarrow +\infty$,

$$S_\varepsilon(m) \sim K_\varepsilon e^{\alpha m} m^{\beta_\varepsilon} \exp\left(\sum_{r \geq 1} \lambda_r^{(\varepsilon)} m^{-r}\right), \quad (2)$$

$$\lambda_r^{(\varepsilon)} = \frac{(-1)^{r+1}}{r(r+1)} \left\{ B_{r+1}\left(\varepsilon + \frac{1}{2}\right) - B_{r+1}(1) \right\}. \quad (3)$$

Writing $\exp(\sum_{r \geq 1} \lambda_r t^r) = \sum_{r \geq 0} c_r t^r$,

$$c_r^{(\varepsilon)} = \frac{1}{r!} Y_r\left(1! \lambda_1^{(\varepsilon)}, 2! \lambda_2^{(\varepsilon)}, \dots, r! \lambda_r^{(\varepsilon)}\right), \quad (4)$$

where Y_r is the complete exponential Bell polynomial.

Let $M_\varepsilon = S_\varepsilon^{-1}$, and set

$$w_0(y) = -\frac{1}{2\alpha} W_{-1}\left(-\frac{2\alpha}{\pi y^2}\right), \quad (5)$$

$$w_1(y) = \frac{1}{2\alpha} W_0\left(\frac{\pi\alpha}{2} y^2\right). \quad (6)$$

Then

$$M_\varepsilon(y) \sim w_\varepsilon(y) + \sum_{r \geq 1} d_r^{(\varepsilon)} w_\varepsilon(y)^{-r}. \quad (7)$$

If $H_\varepsilon(t) = \sum_{r \geq 1} \lambda_r^{(\varepsilon)} t^r$, $P_\varepsilon(t) = \alpha + \beta_\varepsilon t$, and

$$\mathcal{D}_\varepsilon = -\frac{t^2}{P_\varepsilon(t)} \frac{d}{dt}, \quad (8)$$

then the closed coefficient formula is

$$d_r^{(\varepsilon)} = [t^r] \sum_{q=1}^{\lfloor (r+1)/2 \rfloor} \frac{(-1)^q}{q!} \mathcal{D}_\varepsilon^{q-1} \left(\frac{H_\varepsilon(t)^q}{P_\varepsilon(t)} \right). \quad (9)$$

The inverse in the original index is $N_\varepsilon(y) = 2M_\varepsilon(y) + \varepsilon$.

2 Exact structure and the unavoidable parity split

2.1 Equivalent exact definitions

For $n \in \mathbb{N}_0$, define

$$\text{sw}(n) = \frac{n!}{\lfloor n/2 \rfloor!^2}. \quad (10)$$

This is OEIS A056040. Separating parity gives the elementary identities

$$\text{sw}(2m) = \binom{2m}{m}, \quad (11)$$

$$\text{sw}(2m+1) = (2m+1) \binom{2m}{m}. \quad (12)$$

Equivalently,

$$\text{sw}(n) = 2^{n-(n \bmod 2)} \prod_{k=1}^n k^{(-1)^{k+1}}. \quad (13)$$

The first terms are

$$1, 1, 2, 6, 6, 30, 20, 140, 70, 630, 252, 2772, \dots$$

For later analytic work it is convenient to interpolate the two subsequences separately.

Definition 2.1 (Parity branches). For $m \geq 0$ and $\varepsilon \in \{0, 1\}$, set

$$S_\varepsilon(m) = \frac{\Gamma(2m + \varepsilon + 1)}{\Gamma(m+1)^2}. \quad (14)$$

Then $S_\varepsilon(m) = \text{sw}(2m + \varepsilon)$ whenever m is a nonnegative integer.

The gamma duplication formula gives the common normal form

$$S_\varepsilon(m) = K_\varepsilon 4^m \frac{\Gamma(m + \varepsilon + \frac{1}{2})}{\Gamma(m+1)}, \quad K_\varepsilon = \frac{2^\varepsilon}{\sqrt{\pi}}. \quad (15)$$

In particular,

$$S_1(m) = (2m+1)S_0(m). \quad (16)$$

2.2 Generating functions as a consistency check

The exponential generating function follows immediately from Equations (11) and (12):

$$\sum_{n \geq 0} \text{sw}(n) \frac{z^n}{n!} = \sum_{m \geq 0} \frac{z^{2m}}{(m!)^2} + z \sum_{m \geq 0} \frac{z^{2m}}{(m!)^2} \quad (17)$$

$$= (1+z)I_0(2z), \quad (18)$$

where I_0 is the modified Bessel function. The ordinary generating function is

$$\sum_{n \geq 0} \text{sw}(n) z^n = \frac{1}{\sqrt{1-4z^2}} + \frac{z}{(1-4z^2)^{3/2}}. \quad (19)$$

These exact formulae are not needed for the gamma-ratio derivation, but they independently display the two singular amplitudes that later become the powers $m^{-1/2}$ and $m^{1/2}$.

2.3 Why there is no one-valued inverse of the sequence

The adjacent ratios are exact:

$$\frac{\text{sw}(2m+1)}{\text{sw}(2m)} = 2m+1, \quad (20)$$

$$\frac{\text{sw}(2m+2)}{\text{sw}(2m+1)} = \frac{2}{m+1}. \quad (21)$$

Thus the sequence rises steeply from every even term to the following odd term and, for $m \geq 2$, falls from that odd term to the next even term. Consequently there is no eventual monotonicity and no ordinary global inverse $n = n(y)$.

The inverse is intrinsically two-sheeted

Any formula advertised as *the* inverse swing factorial without specifying parity, a smoothing convention, or a generalized-inverse convention is incomplete. The canonical analytic object is the pair

$$M_0 = S_0^{-1}, \quad M_1 = S_1^{-1},$$

or, in the original index variable,

$$N_0(y) = 2M_0(y), \quad N_1(y) = 2M_1(y) + 1.$$

2.4 Monotonicity of each analytic sheet

Proposition 2.2. *The function S_0 is strictly increasing on $(0, \infty)$, with $S_0(0) = 1$, and S_1 is strictly increasing on $[0, \infty)$, with $S_1(0) = 1$. Hence each branch maps $[0, \infty)$ continuously onto $[1, \infty)$ and has a unique real inverse $M_\varepsilon : [1, \infty) \rightarrow [0, \infty)$.*

Proof. Differentiating Equation (14) logarithmically gives

$$\frac{S'_\varepsilon(m)}{S_\varepsilon(m)} = 2\psi(2m + \varepsilon + 1) - 2\psi(m + 1), \quad (22)$$

where $\psi = \Gamma'/\Gamma$. For $x > 0$, $\psi'(x) = \sum_{k \geq 0} (x+k)^{-2} > 0$, so ψ is strictly increasing. If $\varepsilon = 0$, the two arguments in Equation (22) coincide only at $m = 0$ and the first is larger for $m > 0$. If $\varepsilon = 1$, the first is larger for every $m \geq 0$. The range assertion follows from continuity, the initial values, and the exponential growth established below. $\square$

3 The coefficient-calculus toolkit

3.1 Bernoulli polynomials

We use the convention

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} B_n(x) \frac{t^n}{n!}, \quad B_1 = -\frac{1}{2}. \quad (23)$$

The two identities needed repeatedly are

$$B_n\left(\frac{1}{2}\right) = (2^{1-n} - 1)B_n, \quad B_n(1) = B_n \quad (n \neq 1). \quad (24)$$

3.2 Complete Bell polynomials

The complete exponential Bell polynomials Y_n are defined by

$$\exp\left(\sum_{k \geq 1} x_k \frac{t^k}{k!}\right) = \sum_{n \geq 0} Y_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (25)$$

Consequently, if $H(t) = \sum_{k \geq 1} \lambda_k t^k$ and $C(t) = e^{H(t)} = \sum_{n \geq 0} c_n t^n$, then

$$c_n = \frac{1}{n!} Y_n(1! \lambda_1, 2! \lambda_2, \dots, n! \lambda_n), \quad (26)$$

$$= \sum_{\substack{j_1, j_2, \dots \geq 0 \\ j_1 + 2j_2 + \dots = n}} \prod_{k=1}^n \frac{\lambda_k^{j_k}}{j_k!}, \quad (27)$$

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{k=1}^n k \lambda_k c_{n-k}. \quad (28)$$

The first formula is the compact Bell-polynomial form, the second is the frequency-vector or partition form, and the third is the computational recurrence obtained from $C' = H' C$.

3.3 Formal and asymptotic meanings

An identity between coefficient series in the variable t is first interpreted formally. When $t = m^{-1}$ with $m \rightarrow +\infty$, the same algebra gives a Poincaré asymptotic expansion: truncating after t^{N-1} leaves a remainder $\mathcal{O}(t^N)$. This distinction matters because the Bernoulli-generated series below is generally divergent. Formal coefficient identities remain exact even when analytic convergence fails.

Normalization audit

The symbols $B_n(x)$, Y_n , and W_k denote Bernoulli polynomials, complete exponential Bell polynomials, and Lambert-function branches, respectively. Keeping these three unrelated uses of “B/W-type” notation separate prevents the factorial and sign errors that often occur in asymptotic exponentiation and inversion.

4 All-orders direct transseries

4.1 Logarithmic expansion

The shifted Stirling expansion states, for fixed h ,

$$\log \Gamma(m+h) \sim \left(m+h-\frac{1}{2}\right) \log m - m + \frac{1}{2} \log(2\pi) + \sum_{k \geq 2} \frac{(-1)^k B_k(h)}{k(k-1)m^{k-1}}. \quad (29)$$

Subtracting the instances $h = \varepsilon + \frac{1}{2}$ and $h = 1$ in Equation (15) gives the central direct theorem.

Theorem 4.1 (Logarithmic swing-factorial transseries). *For each $\varepsilon \in \{0, 1\}$,*

$$\log S_\varepsilon(m) \sim \log K_\varepsilon + \alpha m + \beta_\varepsilon \log m + \sum_{r \geq 1} \frac{\lambda_r^{(\varepsilon)}}{m^r}, \quad (30)$$

where

$$\lambda_r^{(\varepsilon)} = \frac{(-1)^{r+1}}{r(r+1)} \left[B_{r+1} \left(\varepsilon + \frac{1}{2} \right) - B_{r+1}(1) \right]. \quad (31)$$

For every fixed $N \geq 1$, truncation through $r = N - 1$ has error $\mathcal{O}(m^{-N})$ as $m \rightarrow +\infty$. The same assertion holds uniformly in every closed subsector $|\arg m| \leq \pi - \delta$ that stays away from the gamma poles and chosen logarithmic cuts.

Proof. Insert Equation (29) with $h = \varepsilon + \frac{1}{2}$ and $h = 1$ into

$$\log S_\varepsilon(m) = \log K_\varepsilon + \alpha m + \log \Gamma \left(m + \varepsilon + \frac{1}{2} \right) - \log \Gamma(m + 1).$$

The terms $-m$ and $\frac{1}{2} \log(2\pi)$ cancel. The coefficient of $\log m$ is $(\varepsilon + \frac{1}{2} - \frac{1}{2}) - (1 - \frac{1}{2}) = \varepsilon - \frac{1}{2} = \beta_\varepsilon$. Writing $r = k - 1$ in the remaining Bernoulli sum gives Equation (31). The remainder statement is the corresponding fixed-order remainder theorem for shifted Stirling expansions, applied to a fixed difference of two shifts. $\square$

For the even sheet, Equation (24) simplifies the answer considerably:

$$\lambda_r^{(0)} = \frac{(-1)^{r+1}(2^{-r} - 2)B_{r+1}}{r(r+1)}. \quad (32)$$

Hence

$$\lambda_{2j}^{(0)} = 0 \quad (j \geq 1). \quad (33)$$

The exact branch-transfer identity Equation (16) yields a second useful simplification:

$$\begin{aligned} \log S_1(m) - \log S_0(m) &= \log(2m + 1) = \log 2 + \log m + \log \left(1 + \frac{1}{2m} \right), \\ \lambda_r^{(1)} &= \lambda_r^{(0)} + \frac{(-1)^{r+1}}{r2^r}. \end{aligned} \quad (34)$$

The first logarithmic coefficients are displayed in Table 1.

Table 1: Logarithmic coefficients $\lambda_r^{(\varepsilon)}$.

r	$\lambda_r^{(0)}$	$\lambda_r^{(1)}$
1	$-1/8$	$3/8$
2	0	$-1/8$
3	$1/192$	$3/64$
4	0	$-1/64$
5	$-1/640$	$3/640$
6	0	$-1/384$
7	$17/14336$	$33/14336$
8	0	$-1/2048$
9	$-31/18432$	$-3/2048$
10	0	$-1/10240$

4.2 Multiplicative expansion and general coefficient formulae

Define

$$H_\varepsilon(t) = \sum_{r \geq 1} \lambda_r^{(\varepsilon)} t^r, \quad C_\varepsilon(t) = \exp(H_\varepsilon(t)) = \sum_{r \geq 0} c_r^{(\varepsilon)} t^r. \quad (35)$$

Theorem 4.2 (Multiplicative swing-factorial transseries). *For $\varepsilon \in \{0, 1\}$,*

$$S_\varepsilon(m) \sim K_\varepsilon 4^m m^{\beta_\varepsilon} \sum_{r \geq 0} \frac{c_r^{(\varepsilon)}}{m^r}, \quad (36)$$

where any of the equivalent all-orders formulae

$$c_r^{(\varepsilon)} = \frac{1}{r!} Y_r \left(1! \lambda_1^{(\varepsilon)}, 2! \lambda_2^{(\varepsilon)}, \dots, r! \lambda_r^{(\varepsilon)} \right), \quad (37)$$

$$= \sum_{\substack{j_1, j_2, \dots \geq 0 \\ j_1 + 2j_2 + \dots = r}} \prod_{k=1}^r \frac{(\lambda_k^{(\varepsilon)})^{j_k}}{j_k!}, \quad (38)$$

$$c_0^{(\varepsilon)} = 1, \quad c_r^{(\varepsilon)} = \frac{1}{r} \sum_{k=1}^r k \lambda_k^{(\varepsilon)} c_{r-k}^{(\varepsilon)} \quad (39)$$

compute every coefficient. For fixed N ,

$$S_\varepsilon(m) = K_\varepsilon 4^m m^{\beta_\varepsilon} \left(\sum_{r=0}^{N-1} c_r^{(\varepsilon)} m^{-r} + \mathcal{O}(m^{-N}) \right). \quad (40)$$

Proof. Exponentiate Equation (30) formally and apply Equations (26) to (28). For a fixed truncation order, write the logarithmic remainder as $R_N(m) = \mathcal{O}(m^{-N})$. Then $e^{R_N(m)} = 1 + \mathcal{O}(m^{-N})$, so formal exponentiation also gives the claimed analytic remainder after collecting terms through order $m^{-(N-1)}$. $\square$

Because $S_1 = (2m+1)S_0 = 2m(1+(2m)^{-1})S_0$, the multiplicative coefficients satisfy the particularly simple transfer law

$$\boxed{c_r^{(1)} = c_r^{(0)} + \frac{1}{2} c_{r-1}^{(0)}, \quad c_{-1}^{(0)} = 0.} \quad (41)$$

This identity is exact at the level of formal asymptotic series and is a useful independent check on every computed coefficient.

The first terms are

$$S_0(m) \sim \frac{4^m}{\sqrt{\pi m}} \left(1 - \frac{1}{8m} + \frac{1}{128m^2} + \frac{5}{1024m^3} - \frac{21}{32768m^4} - \frac{399}{262144m^5} + \frac{869}{4194304m^6} + \dots \right), \quad (42)$$

$$S_1(m) \sim 2 \cdot 4^m \sqrt{\frac{m}{\pi}} \left(1 + \frac{3}{8m} - \frac{7}{128m^2} + \frac{9}{1024m^3} + \frac{59}{32768m^4} - \frac{483}{262144m^5} - \frac{2323}{4194304m^6} + \dots \right). \quad (43)$$

Table 2: Multiplicative coefficients $c_r^{(\varepsilon)}$.

r	$c_r^{(0)}$	$c_r^{(1)}$
0	1	1
1	$-1/8$	$3/8$
2	$1/128$	$-7/128$
3	$5/1024$	$9/1024$
4	$-21/32768$	$59/32768$
5	$-399/262144$	$-483/262144$
6	$869/4194304$	$-2323/4194304$
7	$39325/33554432$	$42801/33554432$
8	$-334477/2147483648$	$923923/2147483648$
9	$-28717403/17179869184$	$-30055311/17179869184$
10	$59697183/274877906944$	$-170042041/274877906944$

4.3 A single period-two formula in the original index

For an integer n , let

$$\chi_\varepsilon(n) = \frac{1 + (-1)^{n-\varepsilon}}{2}, \quad \varepsilon \in \{0, 1\}. \quad (44)$$

Then $\chi_\varepsilon(n)$ is the indicator of $n \equiv \varepsilon \pmod{2}$. Since $m = (n - \varepsilon)/2$ on that parity class, Equation (36) becomes the unified period-two transseries

$$\text{sw}(n) \sim \sum_{\varepsilon=0}^1 \chi_\varepsilon(n) \frac{2^n}{\sqrt{\pi}} \left(\frac{n-\varepsilon}{2}\right)^{\varepsilon-1/2} \sum_{r \geq 0} c_r^{(\varepsilon)} \left(\frac{n-\varepsilon}{2}\right)^{-r}. \quad (45)$$

The projectors are not cosmetic: they encode two genuinely different power laws multiplying the common exponential carrier 2^n .

4.4 Late terms, divergence, and the natural exponential scale

The Bernoulli numbers satisfy $B_{2j} = (-1)^{j-1} 2(2j)! \zeta(2j) / (2\pi)^{2j}$. For the nonzero even-sheet logarithmic coefficients this gives

$$\lambda_{2j-1}^{(0)} = \frac{(2^{1-2j} - 2)B_{2j}}{(2j-1)(2j)} \sim (-1)^j \frac{4(2j-2)!}{(2\pi)^{2j}}. \quad (46)$$

The additional term in Equation (34) is merely geometric and therefore does not alter the factorial late-term law on the odd sheet. The logarithmic and multiplicative expansions are consequently divergent Gevrey-type asymptotic series. Their least terms occur at an index proportional to $2\pi m$, and the optimally truncated remainder has the familiar gamma-function scale associated with $e^{-2\pi m}$ in magnitude on the positive real axis. An exact integral that retains this information is given next.

4.5 Exact Binet-integral completion

For $z > 0$, Binet's second formula may be written

$$\mathfrak{L}(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1} dt = \log \Gamma(z). \quad (47)$$

Therefore the exact, untruncated completion of Equation (30) is

$$\log S_\varepsilon(m) = \log K_\varepsilon + \alpha m + \mathfrak{L}\left(m + \varepsilon + \frac{1}{2}\right) - \mathfrak{L}(m + 1). \quad (48)$$

Expanding the elementary part and the arctangent integral in inverse powers of m recovers Equation (31). Retaining the integral instead of expanding it provides a canonical analytic remainder for numerical certification and for hyperasymptotic refinement. In particular, no informal assignment of a Stokes constant is needed to make the real-axis formula exact.

5 Lambert normal forms for the inverse branches

5.1 The dominant phase

Let

$$L_\varepsilon(y) = \log \frac{y}{K_\varepsilon}, \quad F_{0,\varepsilon}(m) = \alpha m + \beta_\varepsilon \log m. \quad (49)$$

The dominant inverse problem is

$$F_{0,\varepsilon}(w) = L_\varepsilon(y), \quad \text{equivalently} \quad K_\varepsilon e^{\alpha w} w^{\beta_\varepsilon} = y. \quad (50)$$

For any nonzero β , exponentiating after division by β gives

$$w = \frac{\beta}{\alpha} W_k \left(\frac{\alpha}{\beta} \left(\frac{y}{K} \right)^{1/\beta} \right). \quad (51)$$

The branch index k must be chosen so that $w \rightarrow +\infty$.

Proposition 5.1 (Real Lambert carriers). *For the even and odd swing-factorial sheets, the unique large positive solutions of Equation (50) are*

$$w_0(y) = -\frac{1}{2\alpha} W_{-1} \left(-\frac{2\alpha}{\pi y^2} \right), \quad (52)$$

$$w_1(y) = \frac{1}{2\alpha} W_0 \left(\frac{\pi\alpha}{2} y^2 \right). \quad (53)$$

Here w_0 is real for sufficiently large y , and w_1 is real for every $y > 0$.

Proof. For $\varepsilon = 0$, $K_0 = \pi^{-1/2}$ and $\beta_0 = -1/2$. Formula Equation (51) becomes

$$w = -\frac{1}{2\alpha} W_k \left(-\frac{2\alpha}{\pi y^2} \right).$$

As $y \rightarrow \infty$, the argument approaches 0^- . The principal branch W_0 tends to zero and gives the small root of the approximate carrier; the branch W_{-1} tends to $-\infty$ and gives the required large positive root. For $\varepsilon = 1$, $K_1 = 2/\sqrt{\pi}$ and $\beta_1 = 1/2$, which gives Equation (53); its argument is positive, so the real branch is W_0 . $\square$

Remark 5.2 (Why two different Lambert branches appear). The even leading carrier $e^{\alpha m} m^{-1/2}$ has a shallow minimum and two real inverse roots for sufficiently large levels if extended down to $m > 0$; the large root is encoded by W_{-1} . The odd carrier $e^{\alpha m} m^{1/2}$ is increasing and uses W_0 . This branch distinction is forced by the sign of β_ε , not by a notational preference.

5.2 The displacement equation

The full logarithmic equation is

$$L_\varepsilon(y) = F_{0,\varepsilon}(m) + H_\varepsilon(m^{-1}), \quad (54)$$

understood asymptotically to arbitrary algebraic order, or exactly after replacing the formal correction by the Binet expression in Equation (48). Since $w = w_\varepsilon(y)$ solves $F_{0,\varepsilon}(w) = L_\varepsilon(y)$, write

$$m = w + D_\varepsilon(t), \quad t = w^{-1}, \quad D_\varepsilon(t) = \sum_{r \geq 1} d_r^{(\varepsilon)} t^r. \quad (55)$$

There is no constant displacement because the omitted logarithmic correction begins at order m^{-1} . Substitution into Equation (54) gives the fundamental formal identity

$$\boxed{\alpha D_\varepsilon(t) + \beta_\varepsilon \log(1 + t D_\varepsilon(t)) + H_\varepsilon\left(\frac{t}{1 + t D_\varepsilon(t)}\right) = 0.} \quad (56)$$

This one equation contains the entire inverse coefficient calculus.

6 All-orders inverse coefficient calculus

6.1 Triangular recurrence

For $r \geq 1$, let

$$D_{\varepsilon, < r}(t) = \sum_{j=1}^{r-1} d_j^{(\varepsilon)} t^j. \quad (57)$$

In the coefficient of t^r in the two nonlinear terms of Equation (56), the unknown $d_r^{(\varepsilon)}$ cannot occur: it is always multiplied by at least one additional factor of t . Thus the recurrence is triangular.

Theorem 6.1 (Triangular inverse recurrence). *The coefficients in Equation (55) are uniquely determined by*

$$\boxed{d_r^{(\varepsilon)} = -\frac{1}{\alpha} [t^r] \beta_\varepsilon \log(1 + t D_{\varepsilon, < r}(t)) + H_\varepsilon\left(\frac{t}{1 + t D_{\varepsilon, < r}(t)}\right).} \quad (58)$$

Only $\lambda_1^{(\varepsilon)}, \dots, \lambda_r^{(\varepsilon)}$ and the previously computed $d_1^{(\varepsilon)}, \dots, d_{r-1}^{(\varepsilon)}$ are needed.

Proof. Take the coefficient of t^r in Equation (56). The linear term contributes $\alpha d_r^{(\varepsilon)}$. Replacing D_ε by $D_{\varepsilon, < r}$ in the remaining terms does not change their t^r coefficient, because the first occurrence of D is through tD . Solving the resulting linear equation proves the formula and uniqueness. $\square$

6.2 A closed Lagrange–Bürmann differential formula

The recurrence is computationally convenient, but a nonrecursive formula reveals the reversion mechanism. Put

$$P_\varepsilon(t) = \alpha + \beta_\varepsilon t, \quad \mathcal{D}_\varepsilon = -\frac{t^2}{P_\varepsilon(t)} \frac{d}{dt}. \quad (59)$$

Since $t = 1/w$, the operator $\mathcal{D}_\varepsilon$ is precisely

$$\mathcal{D}_\varepsilon = \frac{1}{F'_{0,\varepsilon}(w)} \frac{d}{dw}. \quad (60)$$

Lemma 6.2 (Perturbative Lagrange–Bürmann formula). *Let $F'_0(w) \neq 0$, and let $x(q)$ solve*

$$F_0(x(q)) + qh(x(q)) = F_0(w), \quad x(0) = w.$$

Then, as a formal series in q ,

$$x(q) = w + \sum_{n \geq 1} \frac{(-q)^n}{n!} \left(\frac{1}{F'_0(w)} \frac{d}{dw} \right)^{n-1} \left(\frac{h(w)^n}{F'_0(w)} \right). \quad (61)$$

Proof. Use the phase coordinate $u = F_0(x)$, write $x = f(u) = F_0^{-1}(u)$, and set $A(u) = h(f(u))$. The equation becomes the near-identity reversion problem $u + qA(u) = F_0(w)$. Lagrange–Bürmann applied to the observable f gives

$$f(u) = f(R) + \sum_{n \geq 1} \frac{(-q)^n}{n!} \left(\frac{d}{dR} \right)^{n-1} (f'(R) A(R)^n), \quad R = F_0(w).$$

Now $f'(R) = 1/F'_0(w)$ and $d/dR = (1/F'_0(w))d/dw$, which is exactly Equation (61). $\square$

Theorem 6.3 (Closed formula for every inverse coefficient). *For $r \geq 1$,*

$$d_r^{(\varepsilon)} = [t^r] \sum_{q=1}^{\lfloor (r+1)/2 \rfloor} \frac{(-1)^q}{q!} \mathcal{D}_\varepsilon^{q-1} \left(\frac{H_\varepsilon(t)^q}{P_\varepsilon(t)} \right). \quad (62)$$

The sum is finite for each r .

Proof. Apply Theorem 6.2 with $F_0 = F_{0,\varepsilon}$, $h(w) = H_\varepsilon(w^{-1})$, and then set $q = 1$. Formula Equation (60) converts the result into Equation (62). Since $H_\varepsilon(t)^q = \mathcal{O}(t^q)$, while each application of $\mathcal{D}_\varepsilon$ raises the lowest power of t by one, the q -th summand begins at t^{2q-1} . Therefore it can contribute to d_r only if $2q - 1 \leq r$. $\square$

6.3 A completely finite coefficient-array algorithm

Formula Equation (62) contains derivatives, but it can be reduced to finite sums of rational operations. Define

$$h_{q,s}^{(\varepsilon)} = [t^s] H_\varepsilon(t)^q = q! \sum_{\substack{j_1, j_2, \dots \geq 0 \\ \sum_k j_k = q, \sum_k k j_k = s}} \prod_{k \geq 1} \frac{(\lambda_k^{(\varepsilon)})^{j_k}}{j_k!}. \quad (63)$$

For fixed q , introduce coefficients $A_{q,j,r}^{(\varepsilon)}$ by

$$\mathcal{D}_\varepsilon^j \left(\frac{H_\varepsilon(t)^q}{P_\varepsilon(t)} \right) = \sum_{r \geq 0} A_{q,j,r}^{(\varepsilon)} t^r. \quad (64)$$

Since $P_\varepsilon(t)^{-1} = \alpha^{-1} \sum_{v \geq 0} (-\beta_\varepsilon/\alpha)^v t^v$, the initialization and update are

$$A_{q,0,r}^{(\varepsilon)} = \frac{1}{\alpha} \sum_{s=q}^r \left(-\frac{\beta_\varepsilon}{\alpha} \right)^{r-s} h_{q,s}^{(\varepsilon)}, \quad (65)$$

$$A_{q,j+1,r}^{(\varepsilon)} = -\frac{1}{\alpha} \sum_{n=0}^{r-1} n A_{q,j,n}^{(\varepsilon)} \left(-\frac{\beta_\varepsilon}{\alpha} \right)^{r-n-1}. \quad (66)$$

Consequently,

$$d_r^{(\varepsilon)} = \sum_{q=1}^{\lfloor (r+1)/2 \rfloor} \frac{(-1)^q}{q!} A_{q,q-1,r}^{(\varepsilon)}. \quad (67)$$

Equations (63) and (65) to (67) are a finite, nonrecursive-in- D coefficient engine. They are particularly useful for exact rational arithmetic because every $\lambda_r^{(\varepsilon)}$ is rational and the only transcendental symbol is $\alpha = \log 4$.

6.4 Generic low-order coefficients

Suppressing the parity superscript and writing $\beta = \beta_\varepsilon$, $\lambda_j = \lambda_j^{(\varepsilon)}$, the first coefficients generated by any of the three equivalent mechanisms are

$$d_1 = -\frac{\lambda_1}{\alpha}, \quad (68)$$

$$d_2 = -\frac{\lambda_2}{\alpha} + \frac{\beta\lambda_1}{\alpha^2}, \quad (69)$$

$$d_3 = -\frac{\lambda_3}{\alpha} + \frac{\beta\lambda_2 - \lambda_1^2}{\alpha^2} - \frac{\beta^2\lambda_1}{\alpha^3}, \quad (70)$$

$$d_4 = -\frac{\lambda_4}{\alpha} + \frac{\beta\lambda_3 - 3\lambda_1\lambda_2}{\alpha^2} + \frac{-\beta^2\lambda_2 + \frac{5}{2}\beta\lambda_1^2}{\alpha^3} + \frac{\beta^3\lambda_1}{\alpha^4}, \quad (71)$$

$$d_5 = -\frac{\lambda_5}{\alpha} + \frac{\beta\lambda_4 - 4\lambda_1\lambda_3 - 2\lambda_2^2}{\alpha^2} + \frac{-\beta^2\lambda_3 + 7\beta\lambda_1\lambda_2 - 2\lambda_1^3}{\alpha^3} + \frac{\beta^3\lambda_2 - \frac{9}{2}\beta^2\lambda_1^2}{\alpha^4} - \frac{\beta^4\lambda_1}{\alpha^5}. \quad (72)$$

These formulae expose the combinatorial structure: products of logarithmic coefficients record set partitions generated by composition, while powers of β come from repeated differentiation of the logarithmic carrier.

6.5 Specialized coefficients for the two swing sheets

Substituting $\beta_0 = -1/2$ and the even λ_r 's gives

$$d_1^{(0)} = \frac{1}{8\alpha}, \quad d_2^{(0)} = \frac{1}{16\alpha^2}, \quad (73)$$

$$d_3^{(0)} = -\frac{\alpha^2 + 3\alpha - 6}{192\alpha^3}, \quad d_4^{(0)} = -\frac{2\alpha^2 + 15\alpha - 12}{768\alpha^4}, \quad (74)$$

$$d_5^{(0)} = \frac{12\alpha^4 + 20\alpha^3 + 20\alpha^2 - 135\alpha + 60}{7680\alpha^5}, \quad (75)$$

$$d_6^{(0)} = \frac{24\alpha^4 + 90\alpha^3 + 215\alpha^2 - 420\alpha + 120}{30720\alpha^6}. \quad (76)$$

For the odd sheet,

$$d_1^{(1)} = -\frac{3}{8\alpha}, \quad d_2^{(1)} = \frac{2\alpha + 3}{16\alpha^2}, \quad (77)$$

$$d_3^{(1)} = -\frac{3\alpha^2 + 13\alpha + 6}{64\alpha^3}, \quad d_4^{(1)} = \frac{4\alpha^3 + 42\alpha^2 + 53\alpha + 12}{256\alpha^4}, \quad (78)$$

$$d_5^{(1)} = -\frac{12\alpha^4 + 280\alpha^3 + 720\alpha^2 + 445\alpha + 60}{2560\alpha^5}, \quad (79)$$

$$d_6^{(1)} = \frac{80\alpha^5 + 1872\alpha^4 + 9030\alpha^3 + 10845\alpha^2 + 4020\alpha + 360}{30720\alpha^6}. \quad (80)$$

Theorem 6.4 (All-orders inverse transseries). *For every fixed $R \geq 0$, as $y \rightarrow +\infty$,*

$$M_\varepsilon(y) = w_\varepsilon(y) + \sum_{r=1}^R \frac{d_r^{(\varepsilon)}}{w_\varepsilon(y)^r} + \mathcal{O}(w_\varepsilon(y)^{-R-1}), \quad (81)$$

where w_0, w_1 are given by [Equations \(52\) and \(53\)](#). Equivalently, for the original index,

$$N_\varepsilon(y) = 2w_\varepsilon(y) + \varepsilon + 2 \sum_{r=1}^R \frac{d_r^{(\varepsilon)}}{w_\varepsilon(y)^r} + \mathcal{O}(w_\varepsilon(y)^{-R-1}). \quad (82)$$

Proof. The formal construction cancels the residual in [Equation \(54\)](#) through order w^{-R} , leaving $\mathcal{O}(w^{-R-1})$. The derivative of the full logarithmic phase is

$$\alpha + \frac{\beta_\varepsilon}{m} + \mathcal{O}(m^{-2}) = \alpha + o(1),$$

so it is bounded away from zero for large m . The mean-value theorem therefore converts the phase residual into an inverse-variable error of the same order. Finally $N_\varepsilon = 2M_\varepsilon + \varepsilon$. $\square$

Remark 6.5 (Residual certification). A robust implementation should not trust a long symbolic expression merely because it looks patterned. Given a proposed truncation $D_R(t)$, substitute it into the left-hand side of [Equation \(56\)](#); the coefficients of $t, t^2, \dots, t^R$ must vanish identically. This supplies an exact algebraic certificate for every table of d_r 's.

7 Unfolding the Lambert core: polynomial–logarithmic transseries

The Lambert-normalized expansion is usually the most accurate form for computation because it resums the entire dominant power–logarithmic phase. For comparison with classical inverse asymptotics, however, it is useful to expand everything in powers of $1/\log y$ with polynomial coefficients in $\log \log y$.

7.1 Normalized variables

Set

$$x = \alpha m, \quad X_\varepsilon = L_\varepsilon(y) + \beta_\varepsilon \log \alpha = \log \left(\frac{y \alpha^{\beta_\varepsilon}}{K_\varepsilon} \right), \quad \ell_\varepsilon = \log X_\varepsilon. \quad (83)$$

Explicitly,

$$X_0 = \log \left(y \sqrt{\frac{\pi}{\alpha}} \right), \quad X_1 = \log \left(\frac{y \sqrt{\pi \alpha}}{2} \right). \quad (84)$$

Define scaled logarithmic coefficients

$$\mu_k^{(\varepsilon)} = \alpha^k \lambda_k^{(\varepsilon)}. \quad (85)$$

Then [Equation \(54\)](#) becomes

$$X_\varepsilon = x + \beta_\varepsilon \log x + \sum_{k \geq 1} \mu_k^{(\varepsilon)} x^{-k}. \quad (86)$$

We seek

$$x \sim X - \beta \ell + \sum_{r \geq 1} \frac{P_r(\ell)}{X^r}, \quad \ell = \log X, \quad (87)$$

where the parity labels are temporarily suppressed.

7.2 All-orders polynomial recurrence

Let $u = X^{-1}$ and, for $r \geq 1$, define

$$R_{<r}(u, \ell) = -\beta\ell + \sum_{j=1}^{r-1} P_j(\ell)u^j. \quad (88)$$

Substituting $x = X + R$ into Equation (86) gives

$$R + \beta\ell + \beta \log(1 + uR) + \sum_{k \geq 1} \mu_k \left(\frac{u}{1 + uR} \right)^k = 0. \quad (89)$$

Theorem 7.1 (Polynomial–logarithmic coefficient recurrence). *The polynomials $P_r(\ell)$ are uniquely determined by*

$$P_r(\ell) = -[u^r] \beta \log(1 + uR_{<r}(u, \ell)) + \sum_{k=1}^r \mu_k \left(\frac{u}{1 + uR_{<r}(u, \ell)} \right)^k. \quad (90)$$

For every $r \geq 1$, P_r has degree at most r , and because $\beta \neq 0$ in both swing-factorial sheets,

$$[\ell^r] P_r(\ell) = \frac{\beta^{r+1}}{r}. \quad (91)$$

Thus every scalar coefficient $p_{r,j} = [\ell^j] P_r(\ell)$ has the finite general formula obtained by applying $[u^r \ell^j]$ to the right-hand side of Equation (90).

Proof. In the coefficient of u^r in the nonlinear terms of Equation (89), P_r cannot occur because it would enter through uR and hence at order at least u^{r+1} . The u^r coefficient of $R + \beta\ell$ is exactly P_r , proving the recurrence and uniqueness.

Assume inductively that $\deg P_j \leq j$ for $j < r$. Every occurrence of $R_{<r}$ contributes at most one power of ℓ per power of u , so the coefficient of u^r has degree at most r . The terms involving μ_k lose at least $k \geq 1$ powers of the available uR factors and therefore have degree at most $r - 1$. The top-degree coefficient is consequently the same as in the pure equation $X = x + \beta \log x$, whose exact Lambert solution, or a direct induction in Equation (90), gives β^{r+1}/r . $\square$

7.3 Generic first polynomials

The recurrence gives

$$P_1(\ell) = \beta^2\ell - \mu_1, \quad (92)$$

$$P_2(\ell) = \frac{\beta^3}{2}\ell^2 - (\beta^3 + \beta\mu_1)\ell + \beta\mu_1 - \mu_2, \quad (93)$$

$$P_3(\ell) = \frac{\beta^4}{3}\ell^3 - \left(\frac{3}{2}\beta^4 + \beta^2\mu_1 \right) \ell^2 + (\beta^4 + 3\beta^2\mu_1 - 2\beta\mu_2) \ell - \beta^2\mu_1 + \beta\mu_2 - \mu_1^2 - \mu_3, \quad (94)$$

$$P_4(\ell) = \frac{\beta^5}{4}\ell^4 - \left(\frac{11}{6}\beta^5 + \beta^3\mu_1 \right) \ell^3 + \left(3\beta^5 + \frac{11}{2}\beta^3\mu_1 - 3\beta^2\mu_2 \right) \ell^2 + (-\beta^5 - 6\beta^3\mu_1 + 5\beta^2\mu_2 - 3\beta\mu_1^2 - 3\beta\mu_3) \ell + \beta^3\mu_1 - \beta^2\mu_2 + \frac{5}{2}\beta\mu_1^2 + \beta\mu_3 - 3\mu_1\mu_2 - \mu_4. \quad (95)$$

These expressions separate the universal Lambert contribution (the pure powers of β) from the gamma-ratio corrections (the terms containing at least one μ_k).

7.4 Specialized logarithmic expansions

For the even sheet, the first three correction polynomials are

$$P_1^{(0)}(\ell) = \frac{1}{4}\ell + \frac{\alpha}{8}, \quad (96)$$

$$P_2^{(0)}(\ell) = -\frac{1}{16}\ell^2 + \left(\frac{1}{8} - \frac{\alpha}{16}\right)\ell + \frac{\alpha}{16}, \quad (97)$$

$$P_3^{(0)}(\ell) = \frac{1}{48}\ell^3 + \left(\frac{\alpha}{32} - \frac{3}{32}\right)\ell^2 + \left(\frac{1}{16} - \frac{3\alpha}{32}\right)\ell - \frac{\alpha^3}{192} - \frac{\alpha^2}{64} + \frac{\alpha}{32}. \quad (98)$$

Thus

$$M_0(y) \sim \frac{1}{\alpha} \left[X_0 + \frac{1}{2}\ell_0 + \frac{P_1^{(0)}(\ell_0)}{X_0} + \frac{P_2^{(0)}(\ell_0)}{X_0^2} + \frac{P_3^{(0)}(\ell_0)}{X_0^3} + \dots \right]. \quad (99)$$

For the odd sheet,

$$P_1^{(1)}(\ell) = \frac{1}{4}\ell - \frac{3\alpha}{8}, \quad (100)$$

$$P_2^{(1)}(\ell) = \frac{1}{16}\ell^2 - \left(\frac{1}{8} + \frac{3\alpha}{16}\right)\ell + \frac{\alpha^2}{8} + \frac{3\alpha}{16}, \quad (101)$$

$$P_3^{(1)}(\ell) = \frac{1}{48}\ell^3 - \left(\frac{3}{32} + \frac{3\alpha}{32}\right)\ell^2 + \left(\frac{1}{16} + \frac{9\alpha}{32} + \frac{\alpha^2}{8}\right)\ell - \frac{3\alpha}{32} - \frac{13\alpha^2}{64} - \frac{3\alpha^3}{64}, \quad (102)$$

and

$$M_1(y) \sim \frac{1}{\alpha} \left[X_1 - \frac{1}{2}\ell_1 + \frac{P_1^{(1)}(\ell_1)}{X_1} + \frac{P_2^{(1)}(\ell_1)}{X_1^2} + \frac{P_3^{(1)}(\ell_1)}{X_1^3} + \dots \right]. \quad (103)$$

Corollary 7.2 (Leading elementary forms). As $y \rightarrow \infty$,

$$M_0(y) = \frac{1}{\alpha} \left(X_0 + \frac{1}{2} \log X_0 + \mathcal{O}\left(\frac{\log X_0}{X_0}\right) \right), \quad (104)$$

$$M_1(y) = \frac{1}{\alpha} \left(X_1 - \frac{1}{2} \log X_1 + \mathcal{O}\left(\frac{\log X_1}{X_1}\right) \right). \quad (105)$$

The opposite signs of $\frac{1}{2} \log X$ are the inverse-side signature of the direct powers $m^{-1/2}$ and $m^{1/2}$.

7.5 Equivalence of the Lambert and logarithmic normalizations

The two inverse expansions do not contain different information. The Lambert form first solves the carrier exactly and then expands only the small gamma-ratio correction. The polynomial–logarithmic form expands the Lambert carrier itself. More concretely, expanding $w_\varepsilon(y)$ in the scale $X^{-1}\mathbb{R}[\ell]$ and then substituting that expansion into Equation (81) reproduces Equation (90) coefficient by coefficient. Conversely, resumming the subfamily of terms that survives when all μ_k vanish reconstructs the appropriate Lambert branch.

Which inverse expansion should be used?

For numerical work, use the Lambert-normalized series Equation (81); it absorbs the largest logarithmic effects and usually needs fewer terms. For symbolic comparison, scale classification, or interaction with a polynomial–logarithmic transseries algebra, use Equations (99) and (103) and

recurrence Equation (90).

8 Beyond all algebraic orders

The coefficient formulae above completely determine the Poincaré transseries, but the Bernoulli late terms show that this series cannot be summed by taking infinitely many terms at a fixed m . There are two clean ways to retain the missing information.

8.1 Exact completion by Binet's integral

Equation (48) is already an exact completion. For a chosen truncation order N , define the exact logarithmic remainder

$$\mathcal{R}_{\varepsilon,N}(m) = \log S_{\varepsilon}(m) - \log K_{\varepsilon} - \alpha m - \beta_{\varepsilon} \log m - \sum_{r=1}^{N-1} \lambda_r^{(\varepsilon)} m^{-r}. \quad (106)$$

It can be evaluated either from gamma functions or from the difference of the two Binet integrals. For fixed N , $\mathcal{R}_{\varepsilon,N}(m) = \mathcal{O}(m^{-N})$. If N is selected near the least term, the remainder is exponentially small in the standard gamma-function sense. This exact remainder is preferable to an unqualified symbol such as “ $+Ce^{-2\pi m}$ ”, because the most useful exponentially improved representations involve terminant functions and Stokes smoothing rather than a single universal constant C .

8.2 Transport of a general exponentially small sector

The inverse transport mechanism is independent of how the beyond-all-orders remainder is represented.

Theorem 8.1 (Transseries-sector transport through inversion). *Suppose an exact or exponentially improved forward phase is written*

$$Y = \Phi_{\text{alg}}(m) + E(m), \quad (107)$$

where $\Phi'_{\text{alg}}(m) \neq 0$ and E is smaller than the algebraic scale retained in Φ_{alg} . Let m_0 solve $\Phi_{\text{alg}}(m_0) = Y$. Formally in the small sector E ,

$$m = m_0 + \sum_{q \geq 1} \frac{(-1)^q}{q!} \left(\frac{1}{\Phi'_{\text{alg}}(m_0)} \frac{d}{dm_0} \right)^{q-1} \left(\frac{E(m_0)^q}{\Phi'_{\text{alg}}(m_0)} \right). \quad (108)$$

In particular,

$$m = m_0 - \frac{E}{\Phi'} + \frac{EE'}{(\Phi')^2} - \frac{\Phi''E^2}{2(\Phi')^3} + \mathcal{O}(E^3, E^2E', E(E')^2), \quad (109)$$

where the functions on the right are evaluated at m_0 .

Proof. This is Theorem 6.2 with $F_0 = \Phi_{\text{alg}}$, $h = E$, and the coupling parameter set to one after formal expansion. The displayed first terms follow by applying the differential operator once. $\square$

For the swing factorial, $\Phi'_{\text{alg}}(m) = \alpha + \mathcal{O}(m^{-1})$. Therefore the leading transport law is especially simple:

$$\Delta M_{\varepsilon}(y) \sim -\frac{E(w_{\varepsilon}(y))}{\alpha}. \quad (110)$$

If a chosen exponentially improved gamma representation has a sector of magnitude $m^\rho e^{-\kappa m}$, its inverse image has magnitude $w_\varepsilon(y)^\rho e^{-\kappa w_\varepsilon(y)}/\alpha$ at first order, while products in Equation (108) generate the higher multiples of that sector. Since $w_\varepsilon(y) \asymp (\log y)/\alpha$, an exponential in w becomes an additional power-like scale in y ; for example,

$$e^{-2\pi w_\varepsilon(y)} = y^{-2\pi/\alpha} (\log y)^{\mathcal{O}(1)} \times \text{an explicitly expandable constant/logarithmic factor.} \quad (111)$$

This observation explains why hyperasymptotic corrections to the inverse form a genuinely finer transseries grid than the algebraic powers of $1/\log y$.

9 Discrete inverses and index recovery

The analytic inverse $M_\varepsilon(y)$ answers the smooth reversion problem. For an integer-valued inverse one usually wants the first term on a fixed parity sheet that reaches a prescribed level.

Definition 9.1 (Branchwise upper inverse). For $y \geq 1$, define

$$J_\varepsilon(y) = \min\{m \in \mathbb{N}_0 : S_\varepsilon(m) \geq y\}, \quad \nu_\varepsilon(y) = 2J_\varepsilon(y) + \varepsilon. \quad (112)$$

By monotonicity,

$$J_\varepsilon(y) = \lceil M_\varepsilon(y) \rceil, \quad \nu_\varepsilon(y) = 2 \lceil M_\varepsilon(y) \rceil + \varepsilon. \quad (113)$$

This identity includes the case in which y is exactly a branch value, because the ceiling of the corresponding integral inverse is unchanged.

In floating-point computation, one should use the asymptotic expansion only to produce a candidate integer. The final answer is certified by comparing $\log S_\varepsilon(m)$ with $\log y$ at the candidate and its neighbor. This avoids a rounding error when y lies exponentially close to an exact branch value.

The first index of either parity that reaches y is

$$\nu_{\min}(y) = \min\{\nu_0(y), \nu_1(y)\}. \quad (114)$$

Although this is a well-defined generalized inverse, it does not make the original sequence monotone after the first hit. The odd sheet eventually wins by an increasing margin. Indeed, from Equations (104) and (105),

$$N_0(y) - N_1(y) = \frac{2}{\alpha} \log \log y + \frac{2}{\alpha} \log \frac{2}{\alpha} - 1 + o(1), \quad y \rightarrow \infty. \quad (115)$$

Thus large levels are first reached on the odd sheet, roughly $(2/\log 4) \log \log y$ indices before the even sheet catches up.

10 Exact-arithmetic generation and residual tests

All coefficients can be generated without numerical approximation. The following compact SymPy-style code mirrors the mathematical recurrences; α is kept symbolic and Bernoulli values are exact rationals.

Listing 1: Exact generation of direct and inverse coefficients.

```
from sympy import (Symbol, Rational, bernoulli, log,
                  series, expand, simplify)

t = Symbol('t')
```

```

alpha = Symbol('alpha', nonzero=True)

def lam(eps, k):
    a = Rational(eps) + Rational(1, 2)
    return simplify(
        (-1)**(k + 1) * (bernoulli(k + 1, a) - bernoulli(k + 1, 1))
        / (k * (k + 1))
    )

def direct_coefficients(eps, order):
    c = [Rational(1)]
    for r in range(1, order + 1):
        c.append(simplify(sum(k * lam(eps, k) * c[r-k]
                               for k in range(1, r+1)) / r))
    return c

def inverse_coefficients(eps, order):
    beta = Rational(eps) - Rational(1, 2)
    H = sum(lam(eps, k) * t**k for k in range(1, order + 1))
    D = 0
    d = []
    for r in range(1, order + 1):
        u = t / (1 + t*D)
        nonlinear = beta*log(1 + t*D) + H.subs(t, u)
        dr = -series(nonlinear, t, 0, r+1).removeO().expand().coeff(t, r) /
        alpha
        dr = simplify(dr)
        d.append(dr)
        D = expand(D + dr*t**r)

    # Algebraic certificate: all terms through t^r vanish.
    residual = alpha*D + beta*log(1+t*D) + H.subs(t, t/(1+t*D))
    assert series(residual, t, 0, r+1).removeO().expand() == 0
    return d

```

The computational workflow is therefore:

1. generate $\lambda_r^{(\varepsilon)}$ from [Equation \(31\)](#);
2. generate $c_r^{(\varepsilon)}$ from [Equation \(39\)](#);
3. generate $d_r^{(\varepsilon)}$ from [Equation \(58\)](#) or [Equation \(67\)](#);
4. verify the formal residual in [Equation \(56\)](#);
5. for a numerical y , evaluate the appropriate Lambert core and add the desired number of inverse corrections;
6. if a discrete answer is needed, certify neighboring integers using log-gamma evaluations.

11 Numerical validation

The following tables use high-precision gamma and Lambert evaluations. They are not proofs; their role is to expose normalization, sign, or branch errors that formal manipulations might otherwise conceal.

11.1 Direct expansion

Let

$$S_{\varepsilon,R}^{\text{dir}}(m) = K_{\varepsilon} 4^m m^{\beta_{\varepsilon}} \sum_{r=0}^R c_r^{(\varepsilon)} m^{-r}. \quad (116)$$

Table 3 reports the relative error $|S_{\varepsilon,R}^{\text{dir}}/S_{\varepsilon} - 1|$.

Table 3: Relative errors of the direct expansion. Column R includes terms through $c_R m^{-R}$.

ε	m	$R = 0$	$R = 2$	$R = 4$	$R = 8$
0	10	1.2573×10^{-2}	4.8642×10^{-6}	1.5087×10^{-8}	1.6336×10^{-12}
0	50	2.5031×10^{-3}	3.9053×10^{-8}	4.8680×10^{-12}	8.5498×10^{-19}
0	200	6.2519×10^{-4}	6.1033×10^{-10}	4.7561×10^{-15}	3.2645×10^{-24}
1	10	3.5645×10^{-2}	8.6312×10^{-6}	1.8177×10^{-8}	1.7087×10^{-12}
1	50	7.4227×10^{-3}	7.0071×10^{-8}	5.8858×10^{-12}	8.9455×10^{-19}
1	200	1.8701×10^{-3}	1.0977×10^{-9}	5.7556×10^{-15}	3.4163×10^{-24}

11.2 Inverse expansion

Choose $y = S_{\varepsilon}(m)$, so the exact inverse is the displayed m , and define

$$M_{\varepsilon,R}^{\text{inv}}(y) = w_{\varepsilon}(y) + \sum_{r=1}^R d_r^{(\varepsilon)} w_{\varepsilon}(y)^{-r}. \quad (117)$$

Table 4 gives the absolute error $|M_{\varepsilon,R}^{\text{inv}}(S_{\varepsilon}(m)) - m|$.

Table 4: Absolute errors of the Lambert-normalized inverse expansion. The “core” column is $R = 0$.

ε	m	core	$R = 2$	$R = 4$	$R = 8$
0	10	9.3505×10^{-3}	6.0195×10^{-7}	3.4687×10^{-9}	1.6428×10^{-12}
0	50	1.8164×10^{-3}	1.9743×10^{-9}	8.0457×10^{-13}	8.0953×10^{-19}
0	200	4.5166×10^{-4}	2.2503×10^{-11}	7.2653×10^{-16}	3.0590×10^{-24}
1	10	2.5271×10^{-2}	1.5678×10^{-4}	1.9145×10^{-6}	5.0787×10^{-10}
1	50	5.3358×10^{-3}	1.3679×10^{-6}	6.7900×10^{-10}	2.9371×10^{-16}
1	200	1.3478×10^{-3}	2.1720×10^{-8}	6.7553×10^{-13}	1.1441×10^{-21}

The odd-sheet inverse converges more slowly at the same m in these low-order tests because its logarithmic correction begins with $+3/(8m)$, three times the magnitude of the even sheet’s first correction. Nevertheless the all-orders pattern and the predicted powers of w^{-1} are clearly visible.

Optimal truncation

The decreasing errors in these tables do not imply convergence as $R \rightarrow \infty$ at fixed m . The direct Bernoulli series and the induced inverse series are asymptotic. Terms should be stopped near their least magnitude; Binet’s exact integral or a terminant expansion should be used beyond that point.

12 A symmetric centered normal form

The uncentered coordinate m is natural from the sequence definition, but the gamma ratio possesses a better symmetric coordinate in which every odd inverse power disappears from the logarithmic correction.

12.1 Common centering of both parity sheets

Define

$$s_\varepsilon = \frac{2\varepsilon + 1}{4}, \quad x = m + s_\varepsilon. \quad (118)$$

Then $s_0 = 1/4$, $s_1 = 3/4$, and

$$m + \varepsilon + \frac{1}{2} = x + s_\varepsilon, \quad m + 1 = x + 1 - s_\varepsilon, \quad \beta_\varepsilon = 2s_\varepsilon - 1. \quad (119)$$

Moreover,

$$K_\varepsilon 4^{-s_\varepsilon} = \frac{1}{\sqrt{2\pi}} \quad (120)$$

for both values of ε . Hence the exact common centered representation is

$$S_\varepsilon(m) = \frac{e^{\alpha x}}{\sqrt{2\pi}} \frac{\Gamma(x + s_\varepsilon)}{\Gamma(x + 1 - s_\varepsilon)}, \quad x = m + s_\varepsilon. \quad (121)$$

This simultaneously equalizes the leading constant and makes the two gamma shifts complementary.

12.2 Even-power logarithmic series

Applying shifted Stirling in the coordinate x gives

$$\log S_\varepsilon(m) \sim -\frac{1}{2} \log(2\pi) + \alpha x + \beta_\varepsilon \log x + \sum_{r \geq 1} \widehat{\lambda}_r^{(\varepsilon)} x^{-r}, \quad (122)$$

where

$$\widehat{\lambda}_r^{(\varepsilon)} = \frac{(-1)^{r+1}}{r(r+1)} [B_{r+1}(s_\varepsilon) - B_{r+1}(1 - s_\varepsilon)]. \quad (123)$$

The Bernoulli reflection identity

$$B_n(1 - z) = (-1)^n B_n(z) \quad (124)$$

therefore implies

$$\widehat{\lambda}_{2q-1}^{(\varepsilon)} = 0, \quad (125)$$

$$\widehat{\lambda}_{2q}^{(\varepsilon)} = -\frac{B_{2q+1}(s_\varepsilon)}{q(2q+1)}, \quad q \geq 1. \quad (126)$$

Since $s_1 = 1 - s_0$,

$$\widehat{\lambda}_{2q}^{(1)} = -\widehat{\lambda}_{2q}^{(0)}. \quad (127)$$

The first nonzero values are

$$\widehat{\lambda}_2^{(0)} = -\frac{1}{64}, \quad \widehat{\lambda}_4^{(0)} = \frac{5}{2048}, \quad \widehat{\lambda}_6^{(0)} = -\frac{61}{49152}, \quad \widehat{\lambda}_8^{(0)} = \frac{1385}{1048576}, \quad (128)$$

with opposite signs for $\varepsilon = 1$.

Let

$$\hat{H}_\varepsilon(z) = \sum_{q \geq 1} \hat{\lambda}_{2q}^{(\varepsilon)} z^q, \quad e^{\hat{H}_\varepsilon(z)} = \sum_{q \geq 0} \hat{c}_q^{(\varepsilon)} z^q. \quad (129)$$

Then

$$\hat{c}_q^{(\varepsilon)} = \frac{1}{q!} Y_q \left(1! \hat{\lambda}_2^{(\varepsilon)}, 2! \hat{\lambda}_4^{(\varepsilon)}, \dots, q! \hat{\lambda}_{2q}^{(\varepsilon)} \right) \quad (130)$$

and

$$S_\varepsilon(m) \sim \frac{e^{\alpha x} x^{\beta_\varepsilon}}{\sqrt{2\pi}} \sum_{q \geq 0} \hat{c}_q^{(\varepsilon)} x^{-2q}. \quad (131)$$

The first two centered expansions are

$$S_0(m) \sim \frac{e^{\alpha x}}{\sqrt{2\pi x}} \left(1 - \frac{1}{64x^2} + \frac{21}{8192x^4} - \frac{671}{524288x^6} + \frac{180323}{134217728x^8} + \dots \right), \quad x = m + \frac{1}{4}, \quad (132)$$

$$S_1(m) \sim \frac{e^{\alpha x} \sqrt{x}}{\sqrt{2\pi}} \left(1 + \frac{1}{64x^2} - \frac{19}{8192x^4} + \frac{631}{524288x^6} - \frac{174317}{134217728x^8} + \dots \right), \quad x = m + \frac{3}{4}. \quad (133)$$

The centered form is not a new asymptotic function; it is a resummation of the same formal series after a translation. Its advantage is structural sparsity.

12.3 Centered Lambert carriers and inverse expansion

The centered carrier equation is

$$y = \frac{e^{\alpha \hat{w}} \hat{w}^{\beta_\varepsilon}}{\sqrt{2\pi}}. \quad (134)$$

Its large real solutions are

$$\hat{w}_0(y) = -\frac{1}{2\alpha} W_{-1} \left(-\frac{\alpha}{\pi y^2} \right), \quad (135)$$

$$\hat{w}_1(y) = \frac{1}{2\alpha} W_0(4\pi\alpha y^2). \quad (136)$$

Using the same operator calculus as in [Section 6](#), but replacing $H_\varepsilon(t)$ by

$$\hat{H}_\varepsilon^{\text{inv}}(t) = \sum_{q \geq 1} \hat{\lambda}_{2q}^{(\varepsilon)} t^{2q}, \quad (137)$$

produces

$$M_\varepsilon(y) \sim \hat{w}_\varepsilon(y) - s_\varepsilon + \sum_{r \geq 1} \hat{d}_r^{(\varepsilon)} \hat{w}_\varepsilon(y)^{-r}. \quad (138)$$

Every formula [Equations \(58\), \(62\) and \(67\)](#) remains valid after placing hats on H, d, w . Because $\hat{H}$ starts at t^2 ,

$$\hat{d}_1^{(\varepsilon)} = 0, \quad (139)$$

and the first coefficients are

$$\hat{d}_2^{(0)} = \frac{1}{64\alpha}, \quad \hat{d}_3^{(0)} = \frac{1}{128\alpha^2}, \quad \hat{d}_4^{(0)} = -\frac{5\alpha^2 - 8}{2048\alpha^3}, \quad (140)$$

$$\hat{d}_2^{(1)} = -\frac{1}{64\alpha}, \quad \hat{d}_3^{(1)} = \frac{1}{128\alpha^2}, \quad \hat{d}_4^{(1)} = \frac{5\alpha^2 - 8}{2048\alpha^3}. \quad (141)$$

At the next orders,

$$\tilde{d}_5^{(0)} = -\frac{7\alpha^2 - 8}{4096\alpha^4}, \quad \tilde{d}_6^{(0)} = \frac{61\alpha^4 - 57\alpha^2 + 48}{49152\alpha^5}, \quad (142)$$

$$\tilde{d}_5^{(1)} = -\frac{7\alpha^2 - 8}{4096\alpha^4}, \quad \tilde{d}_6^{(1)} = -\frac{61\alpha^4 - 57\alpha^2 + 48}{49152\alpha^5}. \quad (143)$$

Table 5: Absolute inverse errors using only the centered carrier $\hat{w}_\varepsilon - s_\varepsilon$ and using centered corrections through $R = 4$.

ε	m	centered core	centered $R = 4$
0	10	1.1103×10^{-4}	2.6060×10^{-9}
0	50	4.4957×10^{-6}	1.0842×10^{-12}
0	200	2.8158×10^{-7}	1.1092×10^{-15}
1	10	9.4239×10^{-5}	2.9096×10^{-9}
1	50	4.3450×10^{-6}	1.1084×10^{-12}
1	200	2.7917×10^{-7}	1.1154×10^{-15}

The improvement over the uncentered core in Table 4 is substantial: centering has absorbed the entire order- m^{-1} logarithmic correction into the carrier coordinate. The centered and uncentered expansions are formally equivalent, but the centered one is often the preferable computational normal form.

13 A reusable theorem for exponential gamma ratios

The swing factorial is an especially symmetric member of a wider class. The derivation can be packaged once and reused.

Theorem 13.1 (Gamma-ratio transseries and inverse). *Let*

$$F(x) = Ke^{\alpha x} \frac{\Gamma(x+a)}{\Gamma(x+b)}, \quad K > 0, \quad \alpha > 0, \quad (144)$$

with fixed real a, b and $\beta = a - b \neq 0$. On any eventually increasing positive-real branch,

$$F(x) \sim Ke^{\alpha x} x^\beta \exp\left(\sum_{r \geq 1} \lambda_r x^{-r}\right), \quad (145)$$

$$\lambda_r = \frac{(-1)^{r+1}}{r(r+1)} (B_{r+1}(a) - B_{r+1}(b)). \quad (146)$$

The multiplicative coefficients are given by Equations (37) to (39). If

$$w(y) = \frac{\beta}{\alpha} W_k\left(\frac{\alpha}{\beta} \left(\frac{y}{K}\right)^{1/\beta}\right) \quad (147)$$

uses the Lambert branch for which $w \rightarrow +\infty$, then

$$F^{-1}(y) \sim w(y) + \sum_{r \geq 1} d_r w(y)^{-r}, \quad (148)$$

where d_r is given by Equations (58), (62) and (67) with $H(t) = \sum_{r \geq 1} \lambda_r t^r$ and $P(t) = \alpha + \beta t$. The unfolded polynomials are given by Equation (90) with $\mu_r = \alpha^r \lambda_r$.

Proof. The direct formula is the difference of two shifted Stirling expansions. Bell exponentiation gives the multiplicative form. The Lambert formula solves $Ke^{\alpha w}w^\beta = y$. The inverse coefficient formulae follow from [Equation \(56\)](#) and [Theorem 6.2](#); the unfolded recurrence follows from the normalization $x \mapsto \alpha x$. Eventual monotonicity guarantees that the selected real inverse branch is well-defined. $\square$

Remark 13.2 (Optimal centering in the general class). Setting $s = (a + b - 1)/2$ and $X = x + s$ changes the shifts to $(1 + \beta)/2$ and $(1 - \beta)/2$, which sum to one. Bernoulli reflection then eliminates every odd inverse power from the logarithmic gamma-ratio correction. The swing-factorial shifts $s_0 = 1/4$, $s_1 = 3/4$ are precisely this general centering rule.

14 Further consequences

14.1 Derivative and conditioning of the inverse

Let

$$\Phi_\varepsilon(m) = \log S_\varepsilon(m). \quad (149)$$

The inverse-function theorem gives the exact identity

$$M'_\varepsilon(y) = \frac{1}{y\Phi'_\varepsilon(M_\varepsilon(y))} = \frac{1}{2y[\psi(2M_\varepsilon + \varepsilon + 1) - \psi(M_\varepsilon + 1)]}. \quad (150)$$

Differentiating the logarithmic transseries yields

$$\Phi'_\varepsilon(m) \sim \alpha + \beta_\varepsilon m^{-1} - \sum_{r \geq 1} r \lambda_r^{(\varepsilon)} m^{-r-1}. \quad (151)$$

Define

$$\frac{1}{\alpha + \beta t - \sum_{r \geq 1} r \lambda_r t^{r+1}} = \sum_{n \geq 0} g_n t^n. \quad (152)$$

Then

$$g_0 = \frac{1}{\alpha}, \quad (153)$$

$$g_n = -\frac{1}{\alpha} \left(\beta g_{n-1} - \sum_{r=1}^{n-1} r \lambda_r g_{n-r-1} \right), \quad n \geq 1. \quad (154)$$

Consequently,

$$M'_\varepsilon(y) \sim \frac{1}{y} \sum_{n \geq 0} g_n^{(\varepsilon)} M_\varepsilon(y)^{-n}, \quad (155)$$

which becomes a Lambert-normalized expansion after substituting [Equation \(81\)](#). In particular,

$$M'_\varepsilon(y) = \frac{1}{\alpha y} \left(1 - \frac{\beta_\varepsilon}{\alpha M_\varepsilon(y)} + \mathcal{O}(M_\varepsilon(y)^{-2}) \right). \quad (156)$$

Thus the absolute inverse sensitivity decays like $1/(y \log 4)$, while a relative perturbation in y produces an additive perturbation of order one in the branch parameter only after multiplication by $1/\log 4$.

14.2 Complex inverse sheets

For complex y , after fixing branches of powers and logarithms, the carrier equation has the family

$$w_{\varepsilon,k}(y) = \frac{\beta_\varepsilon}{\alpha} W_k \left(\frac{\alpha}{\beta_\varepsilon} \left(\frac{y}{K_\varepsilon} \right)^{1/\beta_\varepsilon} \right), \quad k \in \mathbb{Z}. \quad (157)$$

Whenever a selected branch satisfies $|w_{\varepsilon,k}| \rightarrow \infty$ in a sector avoiding gamma poles and logarithmic cuts, the same coefficients $d_r^{(\varepsilon)}$ give

$$M_{\varepsilon,k}(y) \sim w_{\varepsilon,k}(y) + \sum_{r \geq 1} d_r^{(\varepsilon)} w_{\varepsilon,k}(y)^{-r}. \quad (158)$$

The real inverses studied above are the distinguished choices $k = -1$ for the even carrier approaching the negative cut from the real side and $k = 0$ for the positive odd carrier.

14.3 Structural hierarchy of the coefficient families

The complete calculation can be summarized as a chain of functorial coefficient transformations:

Bernoulli polynomials $\xrightarrow{\text{gamma-ratio difference}} \lambda_r \xrightarrow{\text{exp/Bell}} c_r \xrightarrow{\text{Lagrange-Bürmann}} d_r \xrightarrow{\text{Lambert unfolding}} P_r(\ell). $
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(159)

Every arrow is constructive at arbitrary order, and each has both a closed coefficient-extraction description and a triangular recurrence. This is the precise sense in which the calculation belongs to a combinatorial coefficient calculus rather than to a finite list of hand-expanded Stirling terms.

15 Conclusions

The swing factorial has a richer asymptotic inverse structure than its elementary factorial quotient suggests. The decisive facts are as follows.

1. The sequence is a period-two object. Its even and odd subsequences are the gamma ratios S_0 and S_1 , and the alternating drops rule out a single ordinary inverse.
2. The full direct logarithmic expansion has the Bernoulli-polynomial coefficient [Equation \(31\)](#). Complete Bell polynomials, partition sums, and recurrence [Equation \(39\)](#) give every multiplicative coefficient.
3. The real inverse carriers are exact Lambert functions of different branches: W_{-1} on the even sheet and W_0 on the odd sheet.
4. The inverse correction coefficients admit the triangular recurrence [Equation \(58\)](#), the closed differential formula [Equation \(62\)](#), and the finite array formula [Equation \(67\)](#). These are general formulae at arbitrary order, not merely recursive numerical fits.
5. Unfolding the Lambert core gives the polynomial-logarithmic transseries [Equation \(87\)](#); recurrence [Equation \(90\)](#) computes every coefficient $[\ell^j] P_r$.
6. The symmetric translations $m \mapsto m + 1/4$ and $m \mapsto m + 3/4$ produce a common leading constant $1/\sqrt{2\pi}$, eliminate all odd inverse powers from the logarithmic correction, and yield substantially sharper Lambert carriers.
7. Binet's exact integral supplies a rigorous completion beyond all algebraic orders. Any terminant or other exponentially improved decomposition is transported to the inverse by the same Lagrange-Bürmann operator.

Together these results provide a complete, auditable asymptotic framework for the swing factorial: direct values, continuous inverse sheets, discrete threshold inverses, derivatives, complex continuations, exact coefficient generation, and beyond-all-orders refinement all arise from one coherent calculus.

A Formula sheet

For quick reference, the minimal data needed to reproduce the entire calculation are:

$$\begin{aligned}
 \alpha &= \log 4, & K_\varepsilon &= 2^\varepsilon / \sqrt{\pi}, & \beta_\varepsilon &= \varepsilon - 1/2, \\
 \lambda_r^{(\varepsilon)} &= \frac{(-1)^{r+1}}{r(r+1)} [B_{r+1}(\varepsilon + 1/2) - B_{r+1}(1)], \\
 c_0^{(\varepsilon)} &= 1, & c_r^{(\varepsilon)} &= \frac{1}{r} \sum_{k=1}^r k \lambda_k^{(\varepsilon)} c_{r-k}^{(\varepsilon)}, \\
 w_0(y) &= -\frac{1}{2\alpha} W_{-1} \left(-\frac{2\alpha}{\pi y^2} \right), \\
 w_1(y) &= \frac{1}{2\alpha} W_0 \left(\frac{\pi \alpha y^2}{2} \right), \\
 d_r^{(\varepsilon)} &= -\frac{1}{\alpha} [t^r] \beta_\varepsilon \log(1 + t D_{<r}) + H_\varepsilon \left(\frac{t}{1 + t D_{<r}} \right), \\
 X_\varepsilon &= \log \left(y \alpha^{\beta_\varepsilon} / K_\varepsilon \right), & \mu_r^{(\varepsilon)} &= \alpha^r \lambda_r^{(\varepsilon)}.
 \end{aligned}$$

The centered data are

$$\begin{aligned}
 s_\varepsilon &= (2\varepsilon + 1)/4, & \widehat{K} &= 1/\sqrt{2\pi}, \\
 \widehat{\lambda}_{2q-1}^{(\varepsilon)} &= 0, & \widehat{\lambda}_{2q}^{(\varepsilon)} &= -\frac{B_{2q+1}(s_\varepsilon)}{q(2q+1)}, \\
 \widehat{w}_0(y) &= -\frac{1}{2\alpha} W_{-1} \left(-\frac{\alpha}{\pi y^2} \right), \\
 \widehat{w}_1(y) &= \frac{1}{2\alpha} W_0(4\pi \alpha y^2).
 \end{aligned}$$

B Additional inverse coefficients

For independent implementation checks, the next uncentered coefficients are

$$d_7^{(0)} = -\frac{510\alpha^6 + 539\alpha^5 + 357\alpha^4 - 525\alpha^3 - 4025\alpha^2 + 4200\alpha - 840}{430080\alpha^7}, \quad (160)$$

$$d_8^{(0)} = -\frac{6120\alpha^6 + 14014\alpha^5 + 20454\alpha^4 + 15015\alpha^3 - 101220\alpha^2 + 68040\alpha - 10080}{10321920\alpha^8}, \quad (161)$$

$$d_7^{(1)} = -\frac{1}{430080\alpha^7} \left(990\alpha^6 + 12971\alpha^5 + 108549\alpha^4 + 224175\alpha^3 + 159075\alpha^2 + 39480\alpha + 2520 \right), \quad (162)$$

$$d_8^{(1)} = \frac{1}{3440640\alpha^8} \left(1680\alpha^7 + 59232\alpha^6 + 633122\alpha^5 + 2123226\alpha^4 + 2501205\alpha^3 + 1183140\alpha^2 + 210840\alpha + 10080 \right). \quad (163)$$

For the centered normalization,

$$\widehat{d}_7^{(0)} = \frac{167\alpha^4 - 150\alpha^2 + 96}{196608\alpha^6}, \quad \widehat{d}_7^{(1)} = \frac{167\alpha^4 - 150\alpha^2 + 96}{196608\alpha^6}, \quad (164)$$

$$\widehat{d}_8^{(0)} = -\frac{4155\alpha^6 - 1840\alpha^4 + 1536\alpha^2 - 768}{3145728\alpha^7}, \quad \widehat{d}_8^{(1)} = \frac{4155\alpha^6 - 1840\alpha^4 + 1536\alpha^2 - 768}{3145728\alpha^7}. \quad (165)$$

Substitution into [Equation \(56\)](#), with the corresponding centered or uncentered H , annihilates every coefficient through the displayed order.

C Source and method notes

The exact sequence normalization and parity identities are those recorded for OEIS A056040. The shifted gamma expansion, gamma-ratio asymptotics, Binet integral, and Lambert branches follow standard conventions of the NIST Digital Library of Mathematical Functions. The organizational choice to express exponentiation through Bell polynomials and inversion through coefficient extraction follows the coefficient-calculus and series/transseries drafts in the ProveIt repository. All special swing-factorial coefficient tables in this article were regenerated from the displayed formulae and checked by exact formal residual substitution; the numerical tables were independently checked with high-precision gamma and Lambert evaluations.

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